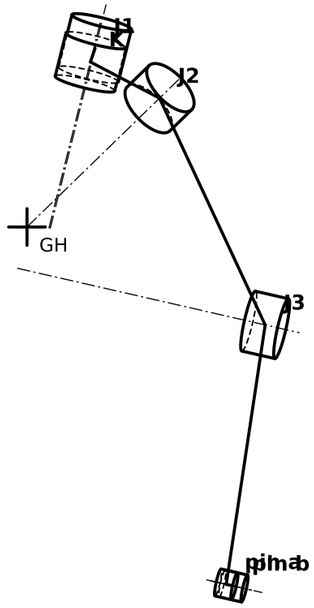
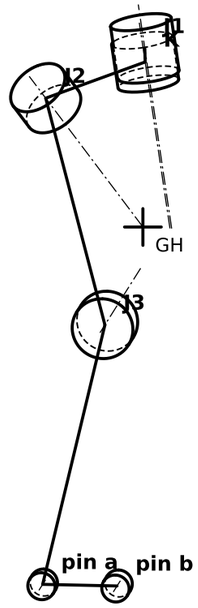
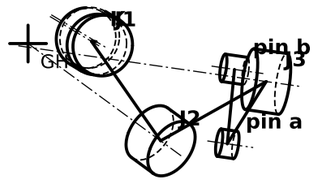
FROM BEHIND (z right, y up)



FROM THE RIGHT SIDE (x right, y up)



FROM ABOVE (z right, x up)



JOINT AXES IN THE BODY FRAME (origin: GH centre; x anterior, y up, z lateral; rest pose; mm)

JOINT	AXIS i, j, k (outward)	ELEV.	AZIM.	REF. POINT x, y, z	FACE FROM GH	AXIS OFF GH	HUB H	∠ TO PREV.	GAP TO PREV.	HUB CTR DIST.
J1	-0.1432 +0.9603 +0.2396	73.79°	120.87°	1.90, 49.68, 20.09	52.24	12.06	17.81			
K	-0.1432 +0.9603 +0.2396	73.79°	120.87°	1.78, 46.76, 18.91	49.17	11.35	14.67	0.00°	0.71	4.69
J2	-0.4654 +0.6168 +0.6347	38.08°	126.25°	-29.10, 38.56, 39.68	62.52	0.00	11.15	35.80°	11.35	41.23
J3	-0.1391 -0.2206 +0.9654	-12.75°	98.20°	-11.61, -30.69, 72.28	78.17	13.84	11.15	57.22°	7.13	83.81
pin a	-0.1391 -0.2206 +0.9654	-12.75°	98.20°	-32.30, -115.89, 62.31	90.22	101.09	5.50	0.00°	87.42	87.90
pin b	-0.1391 -0.2206 +0.9654	-12.75°	98.20°	-7.91, -116.20, 63.69	88.22	99.19	7.50	0.00°	24.35	24.37

REF. POINT: on the axis at the joint's skin-side face (the lower member's face). FACE FROM GH: that face's distance from GH along the axis.
AXIS OFF GH: closest distance from GH to the axis line (0 = the axis passes through the shoulder centre). HUB H: skin-side face to crown.
ELEV.: angle above horizontal. AZIM.: angle in the horizontal plane from anterior (+x) toward lateral (+z).
∠ / GAP TO PREV.: angle and closest distance between this axis and the previous joint's axis (the link between them). HUB CTR DIST.: straight distance between hub mid-points.

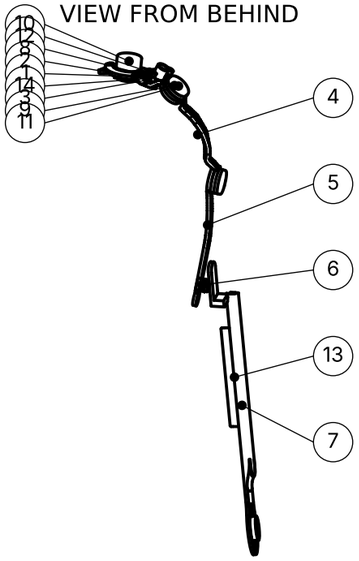
NOTES

1. Use this sheet with the part sheets: each part's END POSITIONS are its slice of this table, in its own frame.
2. J1, K, J2 axes pass through GH (spherical chain); J3 is offset; K is slaved by the cam drive. Knee pins are parallel to J3.
3. Rest pose of the built CAD (arm at the side). Joint angles change with the arm; the part-frame geometry does not.

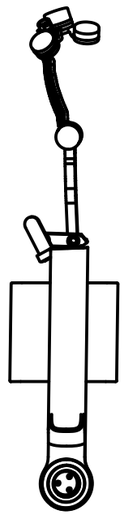
THIRD ANGLE

TITLE: **Cone-cut chain, kinematic layout**		DRAWING NO.: **EXO-A-KIN**
DESIGN: conecut-knee-knuckle-3r (A chain)	PART NO.: **A-KIN**	REV / STATUS: A · for review
MATERIAL: –	MASS: –	QTY PER ASSEMBLY: 1
FINISH: –	SCALE: 1:2	SHEET: 1 / 15
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: joint axes from the part STEP files

VIEW FROM BEHIND



VIEW FROM THE RIGHT SIDE



PARTS LIST

ITEM	PART NO.	DESCRIPTION	QTY	MATERIAL	SOURCE
1	D8-01	Scapula base (J1 upper member)	1	Ti-6Al-4V	made
2	D8-02a	Link 1a (J1 to K, flank lug)	1	Ti-6Al-4V	made
3	D8-02b	Link 1b (K to J2, crank pin)	1	Ti-6Al-4V	made
4	D8-03	Link 2 (J2 to J3)	1	Ti-6Al-4V	made
5	D8-04	Member 3 (J3 disc + E strap)	1	Ti-6Al-4V	made
6	D8-05	Knee link with stop tooth	1	Ti-6Al-4V	made
7	D8-06	Arm pad + pad mount + BOTTOM	1	7075-T6	made (your BOTTOM)
8	D9-01	Drive bell crank	1	Ti-6Al-4V	made
9	D9-02	Drive lobe ring (cam)	1	17-4PH H900	made (wire EDM)
10	H-05	K hub cap with spring chamber	1	Ti-6Al-4V	made
11	H-06	J2 cap (lobe ring seat)	1	Ti-6Al-4V	made
12	H-01..04	Hub cap / spacers / ring nut (sheet HUB)	2+4+4+1	17-4PH	made (turned)
13	D8-07	Upper-arm cuff plate	1	AlSi10Mg	made (DMLS)
14	MR117-2RS	Ball bearing 7 × 11 × 3 (hubs)	8	bearing steel	catalogue
15	MR63ZZ	Ball bearing 3 × 6 × 2.5 (knee 4, follower 1)	5	bearing steel	catalogue
16	683ZZ	Ball bearing 3 × 7 × 3 (crank pivot)	2	bearing steel	catalogue
17	ISO 8734	Dowels: knee pins 3×10, stops 3×6 (3), J3 stop 2×5, nose 2.5×6	7	440C / hardened	catalogue
18	K spring	Clock spring 0.3 × 5 × 145.6, 1.4310	1	spring steel	made / spring maker
19	misc	Pins, shims, e-clips DIN 6799 RS 2.3, wave washer, M4×6 ISO 14042 (6)	lot	stainless	catalogue

NOTES

1. Right shoulder, rest pose. This is design A's 3-joint chain (the current 3-joint lead, verified 93.35). J1, J2, J3 are counted joints; K is slaved by the cam drive (D9-01/02) and not counted.
2. Design A adds a 4th joint, a cuff relay (Ti shell R80.6 strapped 31 mm above the pad, 14 × 14 T-nut carriage, 0-35°, constant-force coil). The relay is NOT modelled in CAD yet: no drawing exists for it.
3. J1 seats on your scapula handler at the acromion top (bracket on the J1 M18×0.5 spigot).
4. Bought parts (items 14-19) are listed, not drawn. Status FOR REVIEW.

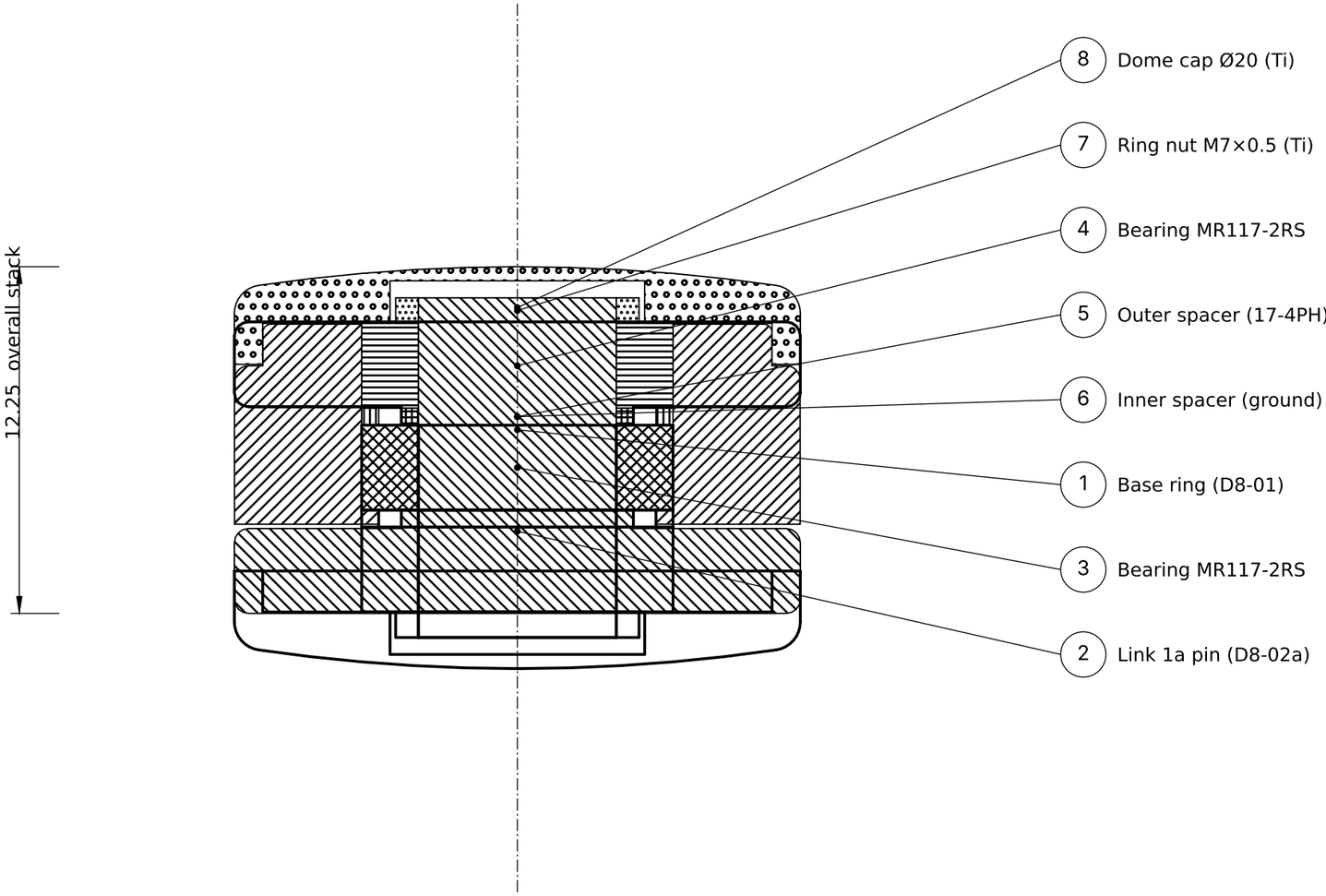
THIRD ANGLE

TITLE: **Cone-cut pin knee chain, general arrangement**		DRAWING NO.: EXO-GA
DESIGN: conecut-knee-knuckle-3r (A chain)	PART NO.: **A-ASSY**	REV / STATUS: A · for review
MATERIAL: see parts list	MASS: 216.1 g	QTY PER ASSEMBLY: 1
FINISH: see part drawings	SCALE: 1:5	SHEET: 2 / 15
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: pipeline/full-exo/designs/user_conecut_k

NOTES

1. Same stack at J1, K, J2 and J3 (K and J2 use their own caps H-05 / H-06): MR117-2RS pair back to back, shaft Ø7 j5 / housing Ø11 J6, preloaded through the ground inner spacer by the M7×0.5 ring nut.
2. 12.25 mm from the skin-side face to the crown (v5 standard, docs/HARDWARE.md).
3. At J1 your scapula handler's bracket threads onto M18×0.5 in place of the dome cap.
4. Drawn from the J1 hub of the built CAD; section plane through the axis, clipped 15 mm around it.

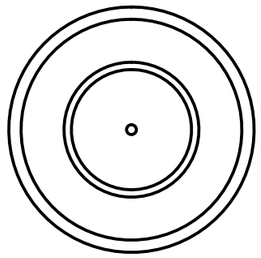
SECTION THROUGH THE HUB AXIS (5:1)



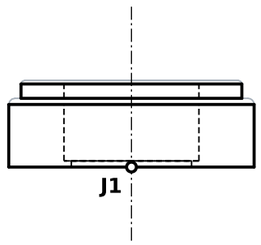
THIRD ANGLE

TITLE **Ø20 hub stack (v5), section**		DRAWING NO. **EXO-A-HUB**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **HUB-v5**	REV / STATUS A · for review
MATERIAL see part list	MASS see parts list	QTY PER ASSEMBLY 4
FINISH bearing seats unblasted	SCALE 5:1	SHEET 3 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE pipeline/full-exo/designs/user_conecut_k

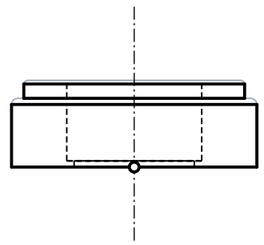
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)

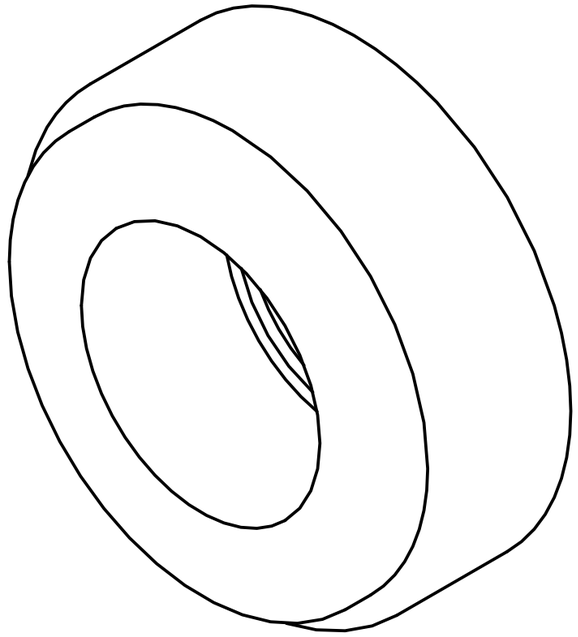


END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J1	0.00	0.00	0.00	0.00°	0.00°	-0.00°	0.00°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

BUILT PART, ISOMETRIC (5:1), for reference



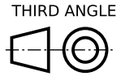
END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J1	Ø20	hub ring OD	pin / boss	0.00	5.10
J1	Ø9.80	hole	bore	0.00	0.50
J1	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
J1	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
J1	faces at s =	0.00, 0.50, 5.60, 7.07			

NOTES

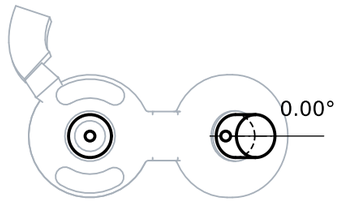
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 61_D8-01_scapula_base_AT_THE_J1_RING.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Your scapula handler's bracket threads onto the M18×0.5 spigot in place of the dome cap; no tab.
5. 5-axis machined.

THIRD ANGLE

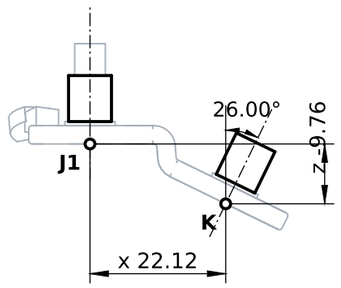


TITLE **Scapula base (J1 upper member)**		DRAWING NO. **EXO-A-D8-01**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **D8-01**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 6.50 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 4 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 61_D8-01_scapula_base_AT_THE_J1_RING

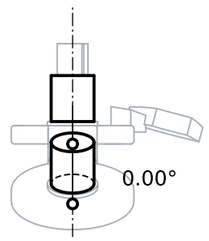
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J1	0.00	0.00	0.00	0.00°	0.00°	0.00°	0.00°
K	22.12	-0.00	-9.76	26.00°	0.00°	26.00°	0.00°

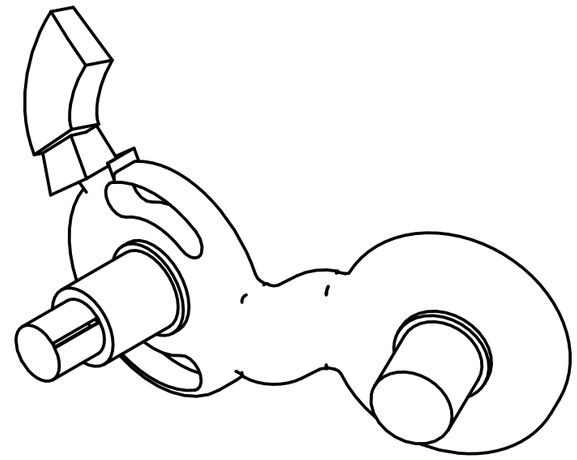
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
J1–K	26.00°	0.00	-55.12	-50.47

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (2:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J1	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
J1	faces at s =	0.00, 1.94, 3.00, 3.65, 11.15, 11.40, 16.30, 16.35			
K	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
K	faces at s =	0.00, 3.00, 3.65, 11.15			

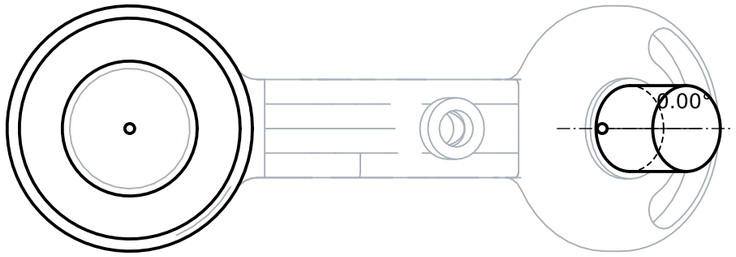
NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 25_D8-02a_link_1a.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. 5-axis machined from solid; R2 centre-line bends (tighter than the 15 mm bent-bar rule).
5. Flank lug: 4-axis UV wire EDM, profile ±0.02, a-C:H DLC 2-3 µm.

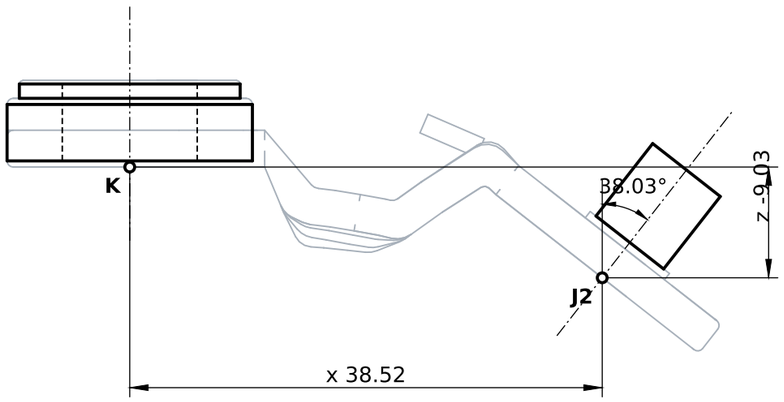
THIRD ANGLE

TITLE **Link 1a (J1 to K, flank lug)**		DRAWING NO. **EXO-A-D8-02a**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **D8-02a**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 12.55 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 1:1	SHEET 5 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 25_D8-02a_link_1a.step

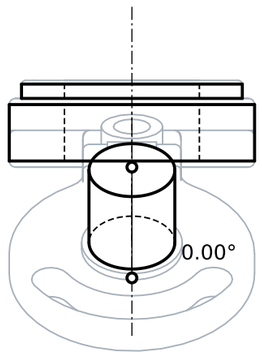
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
K	0.00	0.00	0.00	0.00°	0.00°	0.00°	0.00°
J2	38.52	0.00	-9.03	38.03°	0.00°	38.03°	0.00°

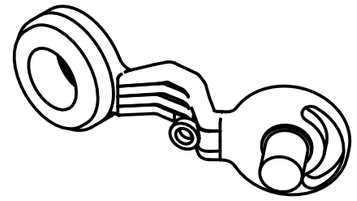
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
K–J2	38.03°	0.00	-58.27	-62.52

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (1:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
K	Ø9.80	hole	bore	0.00	0.50
K	Ø20	hub ring OD	pin / boss	0.50	5.10
K	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
K	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
K	faces at s =	0.00, 0.50, 3.00, 3.50, 5.60, 7.07			
J2	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
J2	faces at s =	0.00, 3.00, 3.65, 11.15			

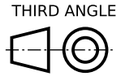
CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
K	Ø3 H7	press, ISO 8734 m6 pin	7.75	12.6°	0.00 – 3.50
K	Ø3 H7	press, ISO 8734 m6 pin	7.75	-167.4°	-0.00 – 3.50
part	Ø3 H7	press, ISO 8734 m6 pin	at 25.8, -0.0, 1.5		len 3.98

NOTES

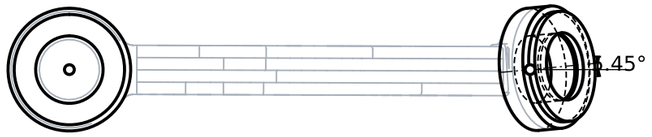
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 26_D8-02b_link_1b.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Machined from solid (3+2 axis).
5. Crank pivot bore Ø3 H7 for the O3 h5/m6 17-4PH pin; J2 pin carries the 118° q2 stop slot.

THIRD ANGLE

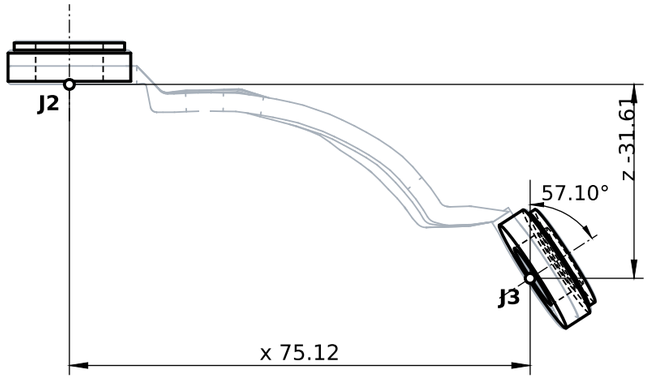


TITLE **Link 1b (K to J2, crank pin)**		DRAWING NO. **EXO-A-D8-02b**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **D8-02b**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 13.93 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 6 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 26_D8-02b_link_1b.step

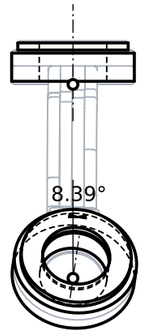
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J2	0.00	0.00	0.00	0.00°	0.00°	-0.00°	0.00°
J3	75.12	0.00	-31.61	57.22°	5.45°	57.10°	8.39°

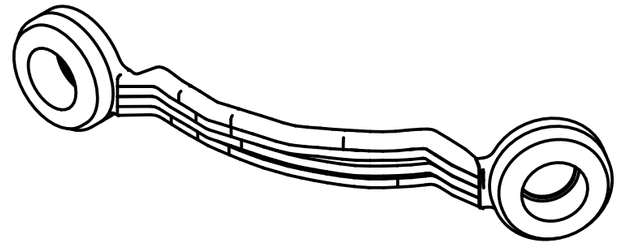
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
J2–J3	57.22°	7.13	-79.77	-88.95

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (1:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J2	Ø9.80	hole	bore	0.00	0.50
J2	Ø20	hub ring OD	pin / boss	0.50	5.10
J2	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
J2	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
J2	faces at s =	0.00, 0.50, 3.00, 3.50, 5.60, 7.07			
J3	Ø9.80	hole	bore	0.00	0.50
J3	Ø20	hub ring OD	pin / boss	0.50	5.10
J3	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
J3	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
J3	faces at s =	0.00, 0.50, 1.00, 3.00, 5.60, 7.07			

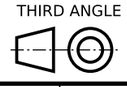
CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
J2	Ø3 H7	press, ISO 8734 m6 pin	7.75	-51.7°	-0.00 – 3.50

NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 30_D8-03_link_2_(5-axis___3+2_machined_from.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. 5-axis / 3+2 machined from solid; 68° drop at J2 (limit 70° on parts machined from solid).
5. J3 stop groove under the J3 ring: 2.10 +0.02/0 wide × 1.00 +0.05/0 deep, centre radius 7.500, 250° arc, full R1.05 ends; concentric to the J3 bore within 0.02.

THIRD ANGLE



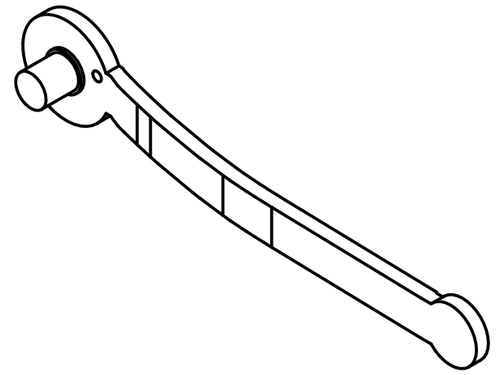
TITLE **Link 2 (J2 to J3, J3 stop groove)**		DRAWING NO. **EXO-A-D8-03**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **D8-03**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 19.66 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 1:1	SHEET 7 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 30_D8-03_link_2_(5-axis___3+2_machine

END POSITIONS (part frame)

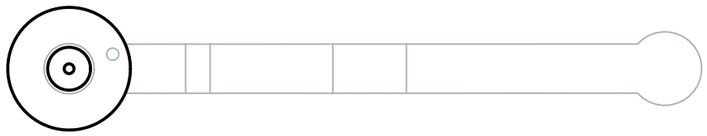
END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J3	0.00	0.00	0.00	0.00°	0.00°	-0.00°	-0.00°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

BUILT PART, ISOMETRIC (1:1), for reference



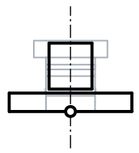
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J3	Ø20	hub ring OD	pin / boss	0.00	3.00
J3	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
J3	faces at s =	0.00, 0.02, 3.00, 3.65, 11.15			

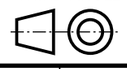
CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
J3	Ø2 −0.005/0	reamed, stop dowel press	7.50	18.4°	0.00 – 3.00

NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 45_D8-04_member_3.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. J3 stop dowel through hole Ø2.000 −0.005/0 reamed at r 7.5 (ISO 8734 2 m6 × 5, ground to 3.90).
5. E boss Ø12 × 2.75 with the Ø3 H7 seat of knee pin a.

THIRD ANGLE

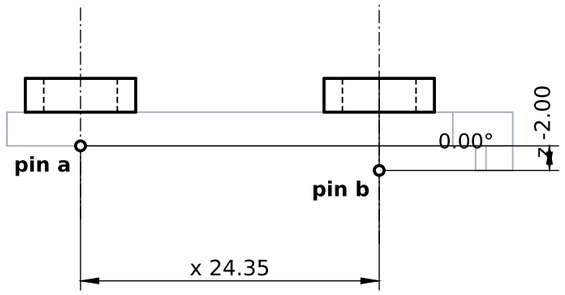


TITLE **Member 3 (J3 disc + E strap)**		DRAWING NO. **EXO-A-D8-04**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **D8-04**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 15.71 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 1:1	SHEET 8 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 45_D8-04_member_3.step

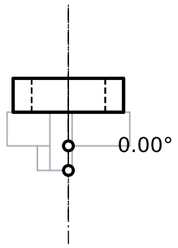
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
pin a	0.00	0.00	0.00	0.00°	0.00°	0.00°	-0.00°
pin b	24.35	0.00	-2.00	0.00°	0.00°	0.00°	-0.00°

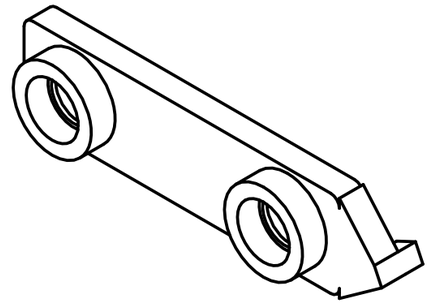
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
pin a–pin b	parallel	24.35	–	–

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (2:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
pin a	Ø6 J6	MR63ZZ housing	bore	0.00	5.50
pin a	Ø5.20	hole	bore	2.50	3.00
pin a	Ø9	bearing boss	pin / boss	2.75	5.50
pin a	faces at s =	2.50, 3.00, 5.50			
pin b	Ø6 J6	MR63ZZ housing	bore	2.00	7.50
pin b	Ø5.20	hole	bore	4.50	5.00
pin b	Ø9	bearing boss	pin / boss	4.75	7.50
pin b	faces at s =	0.00, 2.00, 4.50, 4.75, 5.00, 7.50			

NOTES

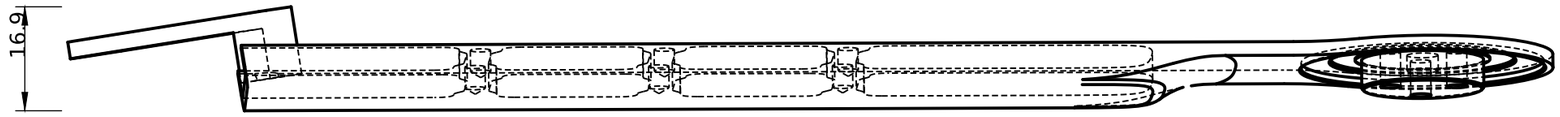
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 52_D8-05_knee_link.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Two Ø9 × 5.5 bosses, each 2 × MR63ZZ in Ø6 J6 housings with a 0.5 mm shoulder.
5. Stop tooth face ±0.02: meets the E strap at knee angle 154.78°.

THIRD ANGLE

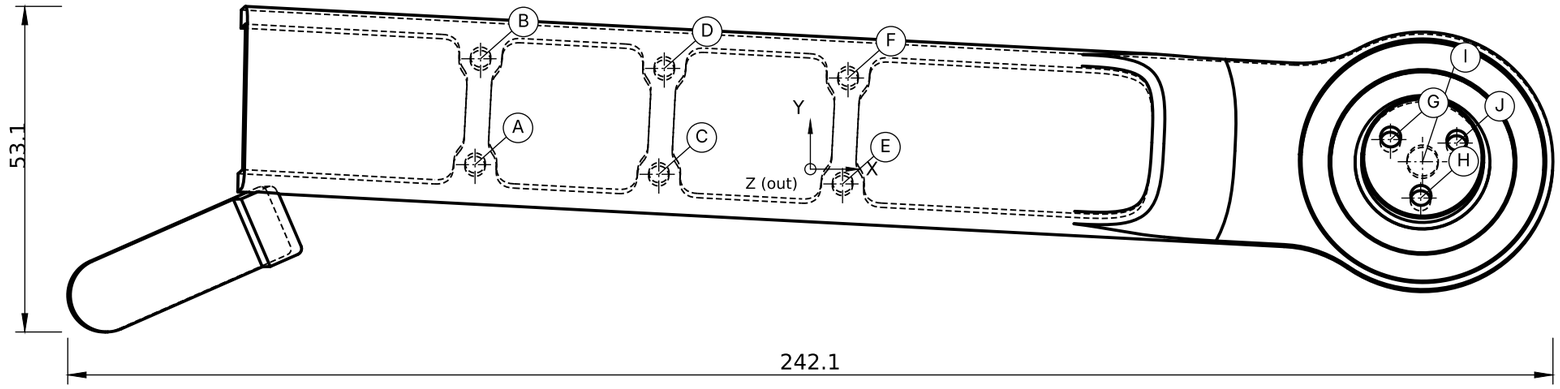


TITLE **Knee link (with stop tooth)**		DRAWING NO. **EXO-A-D8-05**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **D8-05**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 4.98 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 9 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 52_D8-05_knee_link.step

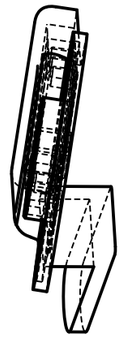
TOP

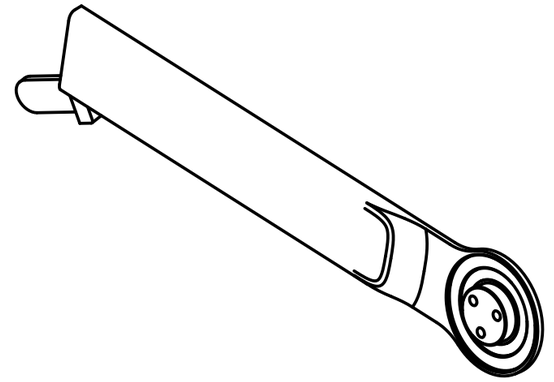


FRONT (looking at the outer face)



RIGHT





ISOMETRIC (1:2)

NOTES

1. Master geometry: STEP file 54_D8-06_BOTTOM_+_arm_pad_+_pad_mount,_one_.step. Dimensions not shown: take from the model.
2. Bores and pins: the hole table. Fits from the design fit list (hub: Ø7 j5 shaft / Ø11 J6 housing for MR117-2RS pairs).
3. Position of tagged features: Ø0.02 to datum A for hub and pin features, Ø0.05 for others (proposed; review before release).
4. Surface: Ra 0.8 on bearing seats and pins, Ra 1.6 on mating faces, Ra 3.2 elsewhere. Break sharp edges 0.2 max.
5. Your BOTTOM bar, cut square to the arm 19.8 mm distal of pin b and merged with the pad mount; the only one of your parts that is modified.
6. Ø3 H7 seat of knee pin b.

HOLE AND PIN TABLE

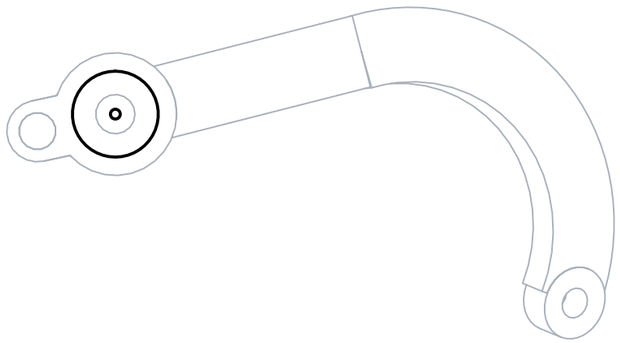
Origin: tag A axis, mid-length (datum A). X, Y, Z: part axes as drawn. Centres in mm.

TAG	SIZE / FIT	FUNCTION	TYPE	LEN	X	Y	Z	AXIS i, j, k
A	Ø3.30	hole	bore	3.50	-54.67	0.73	3.07	+0.012 +0.154 +0.988
B	Ø3.30	hole	bore	3.50	-53.77	18.00	0.36	+0.012 +0.154 +0.988
C	Ø3.30	hole	bore	3.50	-24.71	-0.84	2.96	+0.012 +0.154 +0.988
D	Ø3.30	hole	bore	3.50	-23.81	16.43	0.25	+0.012 +0.154 +0.988
E	Ø3.30	hole	bore	3.50	5.25	-2.41	2.86	+0.012 +0.154 +0.988
F	Ø3.30	hole	bore	3.50	6.14	14.86	0.15	+0.012 +0.154 +0.988
G	Ø3.30	hole	bore	5.60	94.58	4.82	1.74	-0.012 -0.154 -0.988
H	Ø3.30	hole	bore	5.60	99.51	-4.71	3.18	-0.012 -0.154 -0.988
I	Ø5.10	hole	bore	2.50	99.81	1.22	0.68	-0.012 -0.154 -0.988
J	Ø3.30	hole	bore	5.60	105.39	4.26	1.70	-0.012 -0.154 -0.988

THIRD ANGLE

TITLE: **Arm pad + pad mount + BOTTOM (one part)**		DRAWING NO.: **D8-A-D8-06**
DESIGN: conecut-knee-knuckle-3r (A chain)	PART NO.: **D8-06**	REV / STATUS: A · for review
MATERIAL: 7075-T6	MASS: 58.66 g	QTY PER ASSEMBLY: 1
FINISH: clear anodise (never black)	SCALE: 1:1	SHEET: 10 / 15
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: 54_D8-06_BOTTOM_+_arm_pad_+_pad_m

TOP (X right, Y up)

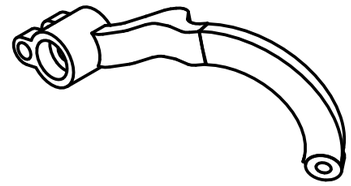


END POSITIONS (part frame)

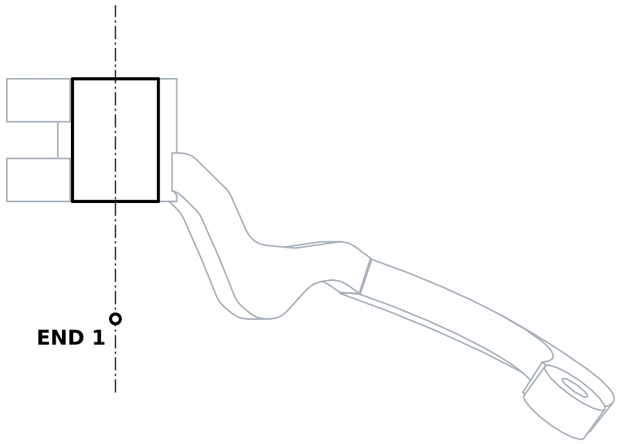
END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
END 1	0.00	0.00	0.00	0.00°	0.00°	-0.00°	0.00°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

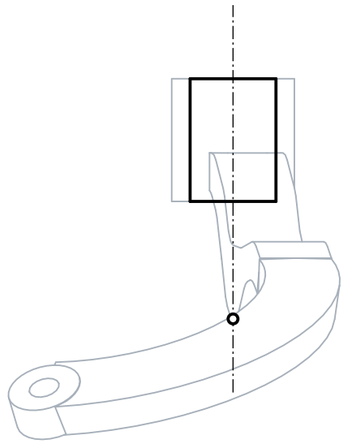
BUILT PART, ISOMETRIC (1:1), for reference



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
END 1	Ø7 H7	bore	bore	9.57	19.57
END 1	Ø3.20	hole	bore	12.57	16.57
END 1	faces at s =	0.00, 9.57, 12.57, 13.07, 13.52, 16.07, 16.57, 19.57			

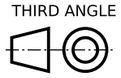
CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
END 1	Ø3 H7	press, ISO 8734 m6 pin	6.50	-167.7°	9.57 – 19.57
part	Ø2.5 −0.005/0	reamed, nose dowel press	at 36.6, -15.0, -6.8		len 3.01

NOTES

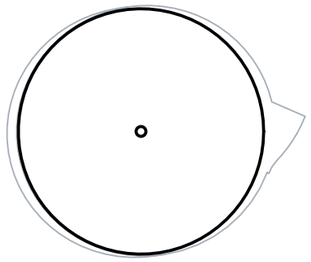
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 32_D9-01_drive_bell_crank_(Ti-6Al-4V,_5-axi.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. 5-axis machined from solid; all bores in one setup ±5 µm.
5. Straddle boss Ø10 × 10 with 683ZZ seats Ø7 J6 at both ends; Ø6 tip boss reamed Ø2.495-2.500 for the nose dowel (Loctite 648).

THIRD ANGLE

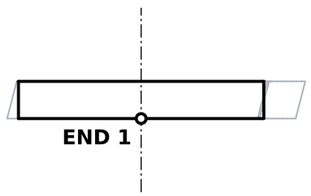


TITLE **Drive bell crank**		DRAWING NO. **EXO-A-D9-01**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **D9-01**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 7.38 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 11 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 32_D9-01_drive_bell_crank_(Ti-6Al-4V,_5-

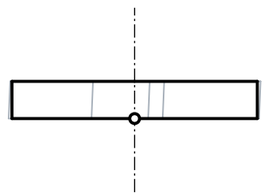
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)

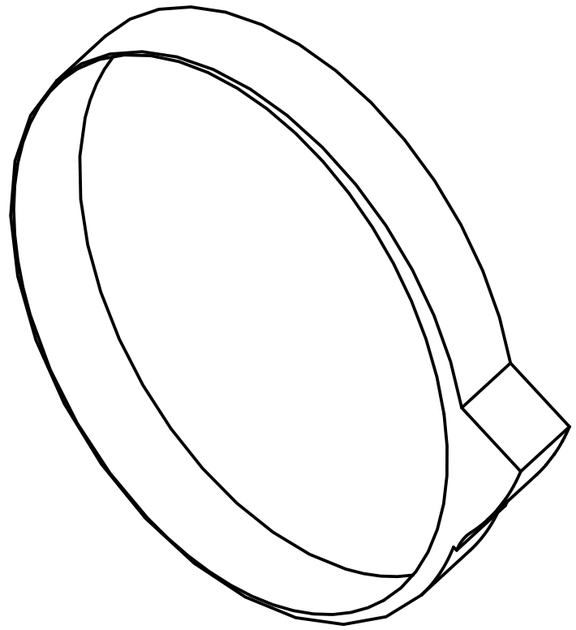


END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
END 1	0.00	0.00	0.00	0.00°	0.00°	0.00°	0.00°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

BUILT PART, ISOMETRIC (5:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
END 1	Ø20 H7	on the cap skirt	bore	0.00	3.03
END 1	faces at s =	0.00, 3.03			

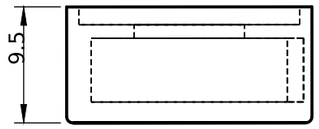
NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 31_D9-02_drive_lobe_ring.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Wire EDM, taper 14.7° along the follower axis; profile r 10.65 → 12.62 (6 three-point arcs), dwell r 10.20.
5. Bore Ø20 H7 on the J2 cap skirt (Ø20 g6), keyed by a Ø1 dowel at the onset phase + 0.35° of q2.

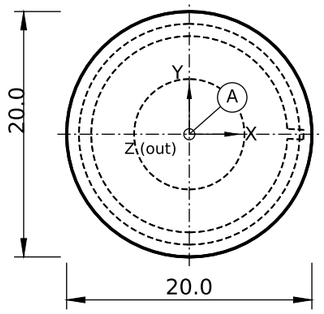
THIRD ANGLE

TITLE **Drive lobe ring (cam)**		DRAWING NO. **EXO-A-D9-02**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **D9-02**	REV / STATUS A · for review
MATERIAL 17-4PH H900	MASS 0.73 g	QTY PER ASSEMBLY 1
FINISH as wire-EDM, passivate	SCALE 2:1	SHEET 12 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 31_D9-02_drive_lobe_ring.step

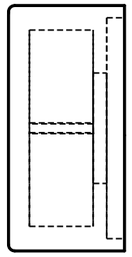
TOP

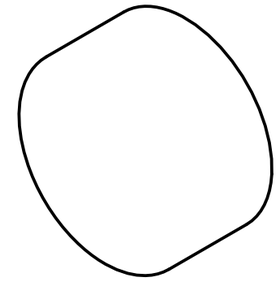


FRONT (looking at the outer face)



RIGHT





ISOMETRIC (2:1)

NOTES

1. Master geometry: STEP file 11_K_clock-spring_cap_O20_x_crown_F+18.25_(.step. Dimensions not shown: take from the model.
2. Bores and pins: the hole table. Fits from the design fit list (hub: Ø7 j5 shaft / Ø11 J6 housing for MR117-2RS pairs).
3. Position of tagged features: Ø0.02 to datum A for hub and pin features, Ø0.05 for others (proposed; review before release).
4. Surface: Ra 0.8 on bearing seats and pins, Ra 1.6 on mating faces, Ra 3.2 elsewhere. Break sharp edges 0.2 max.
5. Crown F+18.25 (6 mm taller than the v5 cap), spring chamber Ø16 × 5.2 with the outer-hook slot, M18×0.5.
6. Pinned to link 1b's ring at the spring's preload index.

HOLE AND PIN TABLE

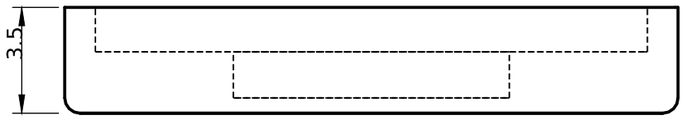
Origin: tag A axis, mid-length (datum A). X, Y, Z: part axes as drawn. Centres in mm.

TAG	SIZE / FIT	FUNCTION	TYPE	LEN	X	Y	Z	AXIS i, j, k
A1	Ø9.00	hole	bore	1.10	-0.00	-0.00	-2.47	-0.001 -0.000 -1.000
A2	Ø16.00	hole	bore	5.20	0.00	0.00	0.67	-0.001 -0.000 -1.000
A3	Ø18.00	hole	bore	1.45	-0.00	-0.00	-3.75	-0.001 -0.000 -1.000
A4	Ø20	hub ring OD	pin	8.95	0.00	0.00	0.00	+0.001 +0.000 +1.000

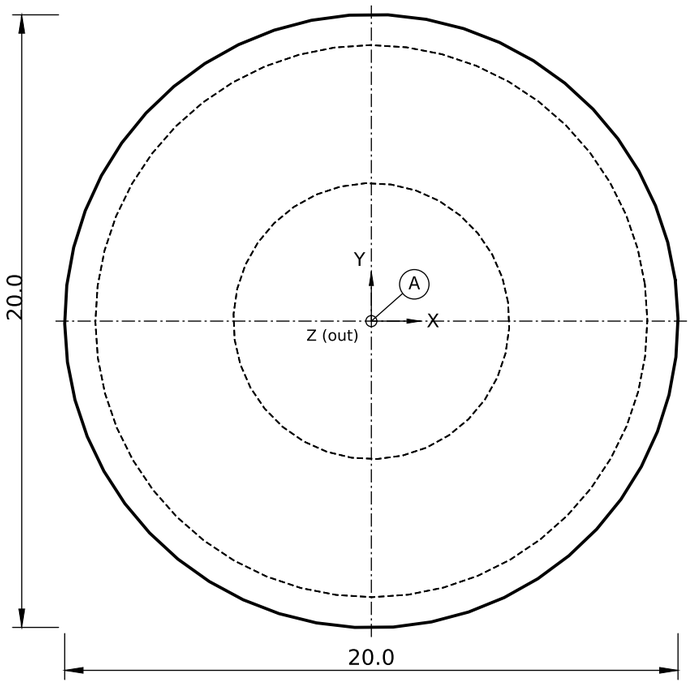
THIRD ANGLE

TITLE **K hub cap with spring chamber**		DRAWING NO. **EXO-A-H-05**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **H-05**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 6.54 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 13 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 11_K_clock-spring_cap_O20_x_crown_F+

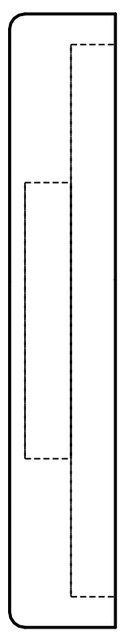
TOP

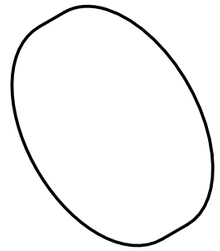


FRONT (looking at the outer face)



RIGHT





ISOMETRIC (2:1)

NOTES

1. Master geometry: STEP file 18_J2_cap_O20_x_crown_F+12.25,_flat_crown,_.step. Dimensions not shown: take from the model.
2. Bores and pins: the hole table. Fits from the design fit list (hub: Ø7 j5 shaft / Ø11 J6 housing for MR117-2RS pairs).
3. Position of tagged features: Ø0.02 to datum A for hub and pin features, Ø0.05 for others (proposed; review before release).
4. Surface: Ra 0.8 on bearing seats and pins, Ra 1.6 on mating faces, Ra 3.2 elsewhere. Break sharp edges 0.2 max.
5. Crown F+12.25, flat, R0.5 corner; skirt Ø20 g6 carries the lobe ring (D9-02). M18×0.5.

HOLE AND PIN TABLE

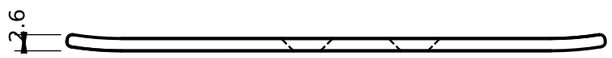
Origin: tag A axis, mid-length (datum A). X, Y, Z: part axes as drawn. Centres in mm.

TAG	SIZE / FIT	FUNCTION	TYPE	LEN	X	Y	Z	AXIS i, j, k
A1	Ø9.00	hole	bore	1.50	0.00	-0.00	0.73	+0.000 -0.000 -1.000
A2	Ø18.00	hole	bore	1.45	0.00	-0.00	-0.75	+0.000 -0.000 -1.000
A3	Ø20	hub ring OD	pin	2.95	0.00	0.00	0.00	-0.000 +0.000 +1.000

THIRD ANGLE

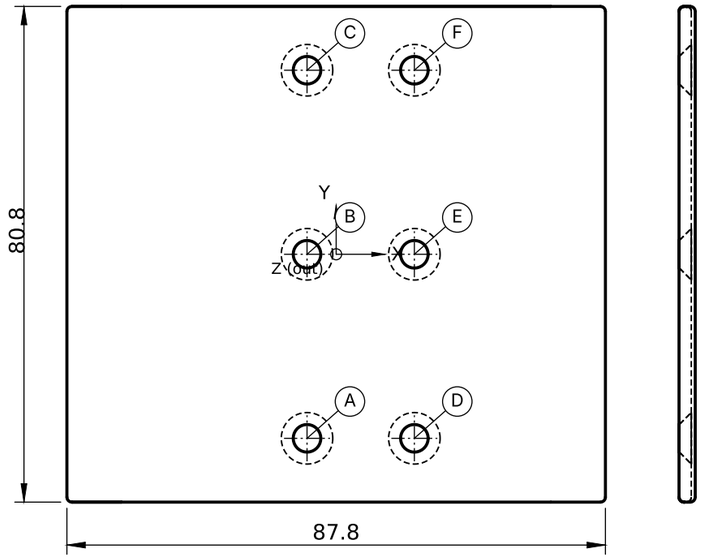
TITLE **J2 cap (lobe ring seat)**		DRAWING NO. **EXO-A-H-06**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **H-06**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 2.73 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 5:1	SHEET 14 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 18_J2_cap_O20_x_crown_F+12.25,_flat_c

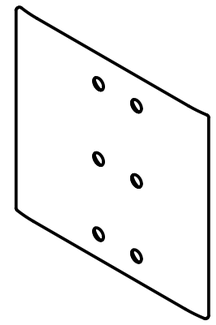
TOP



RIGHT

FRONT (looking at the outer face)





ISOMETRIC (1:2)

NOTES

1. Master geometry: STEP file 99_UPPER_ARM_CUFF_(one_sketch.step. Dimensions not shown: take from the model.
2. Bores and pins: the hole table. Fits from the design fit list (hub: Ø7 j5 shaft / Ø11 J6 housing for MR117-2RS pairs).
3. Position of tagged features: Ø0.02 to datum A for hub and pin features, Ø0.05 for others (proposed; review before release).
4. Surface: Ra 0.8 on bearing seats and pins, Ra 1.6 on mating faces, Ra 3.2 elsewhere. Break sharp edges 0.2 max.
5. Fixed to BOTTOM with 6 × M4×6 ISO 10642.

HOLE AND PIN TABLE

Origin: tag A axis, mid-length (datum A). X, Y, Z: part axes as drawn. Centres in mm.

TAG	SIZE / FIT	FUNCTION	TYPE	LEN	X	Y	Z	AXIS i, j, k
A	Ø4.50	hole	bore	0.05	-4.75	-30.00	1.29	+0.000 +0.000 +1.000
B	Ø4.50	hole	bore	0.05	-4.75	-0.00	1.29	+0.000 +0.000 +1.000
C	Ø4.50	hole	bore	0.05	-4.75	30.00	1.29	+0.000 +0.000 +1.000
D	Ø4.50	hole	bore	0.05	12.75	-30.00	1.29	+0.000 +0.000 +1.000
E	Ø4.50	hole	bore	0.05	12.75	0.00	1.29	+0.000 +0.000 +1.000
F	Ø4.50	hole	bore	0.05	12.75	30.00	1.29	+0.000 +0.000 +1.000

THIRD ANGLE

TITLE **Upper-arm cuff plate**		DRAWING NO. **EXO-A-D8-07**
DESIGN conecut-knee-knuckle-3r (A chain)	PART NO. **D8-07**	REV / STATUS A · for review
MATERIAL AlSi10Mg (DMLS)	MASS 36.57 g	QTY PER ASSEMBLY 1
FINISH bead blast	SCALE 1:1	SHEET 15 / 15
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 99_UPPER_ARM_CUFF_(one_sketch.step